Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Further Integration

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find $\int \frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x$. **[2]**

2. Find $\int \frac{1}{a^2+x^2}\,\mathrm{d}x$. **[2]**

3. Find $\int \frac{1}{\sqrt{x^2+a^2}}\,\mathrm{d}x$. **[3]**

4. Find $\int \frac{1}{\sqrt{x^2-a^2}}\,\mathrm{d}x$. **[3]**

5. Find the volume of revolution when $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. **[4]**

6. Find the volume of revolution when $y = \sqrt{x}$ from $x = 1$ to $x = 4$ is rotated about the x-axis. **[4]**

7. Find $\int \sin^2 x\,\mathrm{d}x$. **[3]**

8. Find $\int \cos^2 x\,\mathrm{d}x$. **[3]**

9. Find $\int \sin^3 x\,\mathrm{d}x$. **[4]**

10. Use integration by parts to find $\int xe^x\,\mathrm{d}x$. **[3]**

11. Find $\int x \sin x \, \mathrm{d}x$. **[3]**

12. Find $\int \frac{x}{x^2+1} \, \mathrm{d}x$. **[2]**

13. Find the arc length of $y = x^{3/2}$ from $x = 0$ to $x = 1$. **[5]**

14. Evaluate $\int_0^1 \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x$. **[3]**

15. Evaluate $\int_0^\infty e^{-x} \, \mathrm{d}x$. **[3]**

Section B – Hard (Q16–25)

16. Find the volume generated when the region under $y = \sin x$ from 0 to π is rotated about the x-axis. **[5]**

17. Find $\int \frac{1}{(x^2+1)^2} \, \mathrm{d}x$ using a trigonometric substitution. **[5]**

18. Find the surface area of revolution of $y = x^2$ from 0 to 1 about the x-axis. **[6]**

19. Evaluate the improper integral $\int_1^\infty \frac{1}{x^2} \, \mathrm{d}x$. **[3]**

20. Find $\int e^{ax} \cos bx \, \mathrm{d}x$. **[5]**

21. Use reduction formula ideas to find $\int \sin^4 x \, \mathrm{d}x$. **[5]**

22. Find the centroid of the region under $y = \sqrt{x}$ from 0 to 4. **[6]**

23. Find $\int \frac{x^2}{\sqrt{a^2-x^2}}\,\mathrm{d}x$. **[5]**

24. Evaluate $\int_0^{\pi/2} \sin^2 x \cos^2 x\,\mathrm{d}x$. **[4]**

25. Find the arc length of the curve $x = t^2$, $y = t^3$ from $t = 0$ to $t = 1$. **[5]**

End of Worksheet • *Show all working clearly.*